

RCO AI Executive Analytics Summary

Healthcare revenue cycle data engineering, inferential statistics, and operational intelligence

Research question: To what extent does authorization turnaround time affect denied charges in healthcare revenue cycle operations?

108,000

Raw operational records

3,000

Daily observations

0.861

Pearson correlation

0.742

R-squared

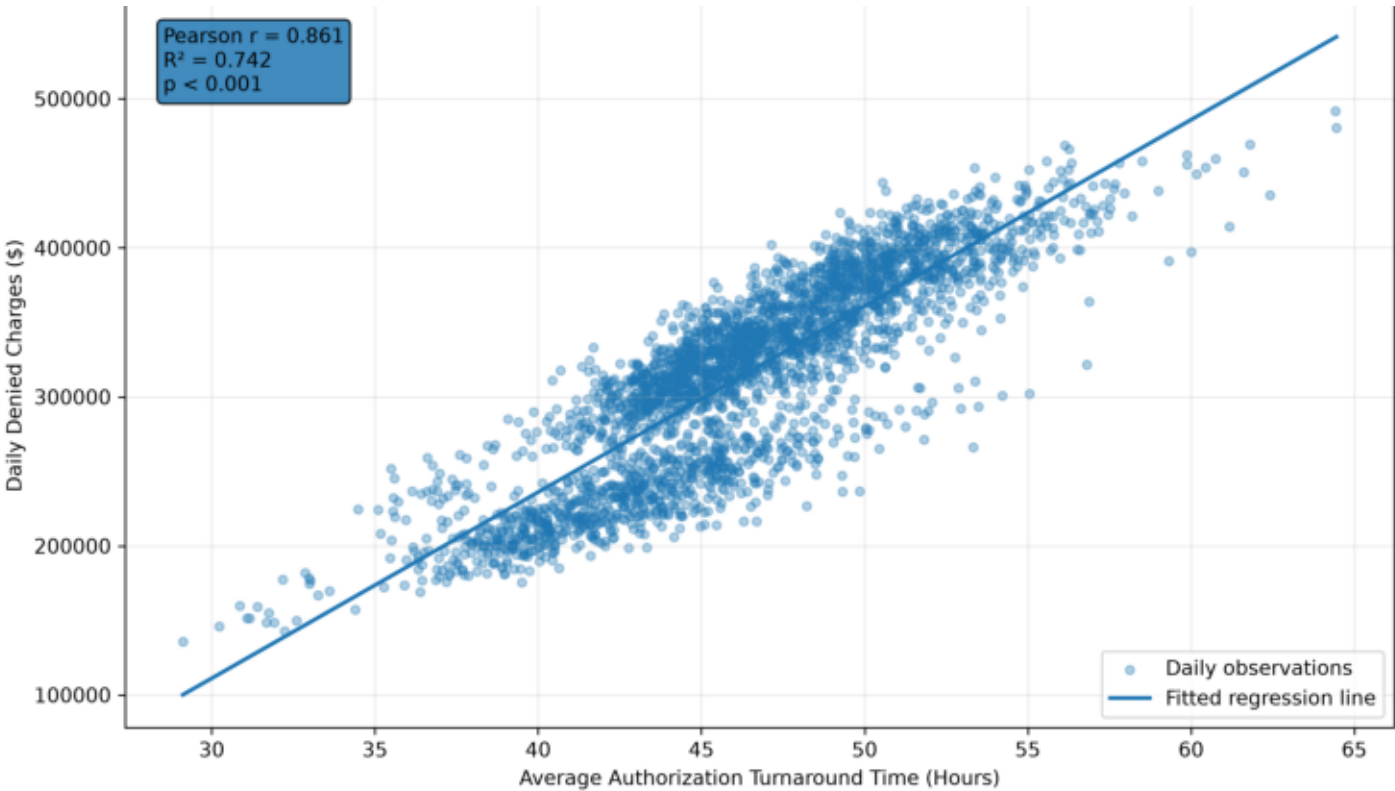
< 0.001

Regression p-value

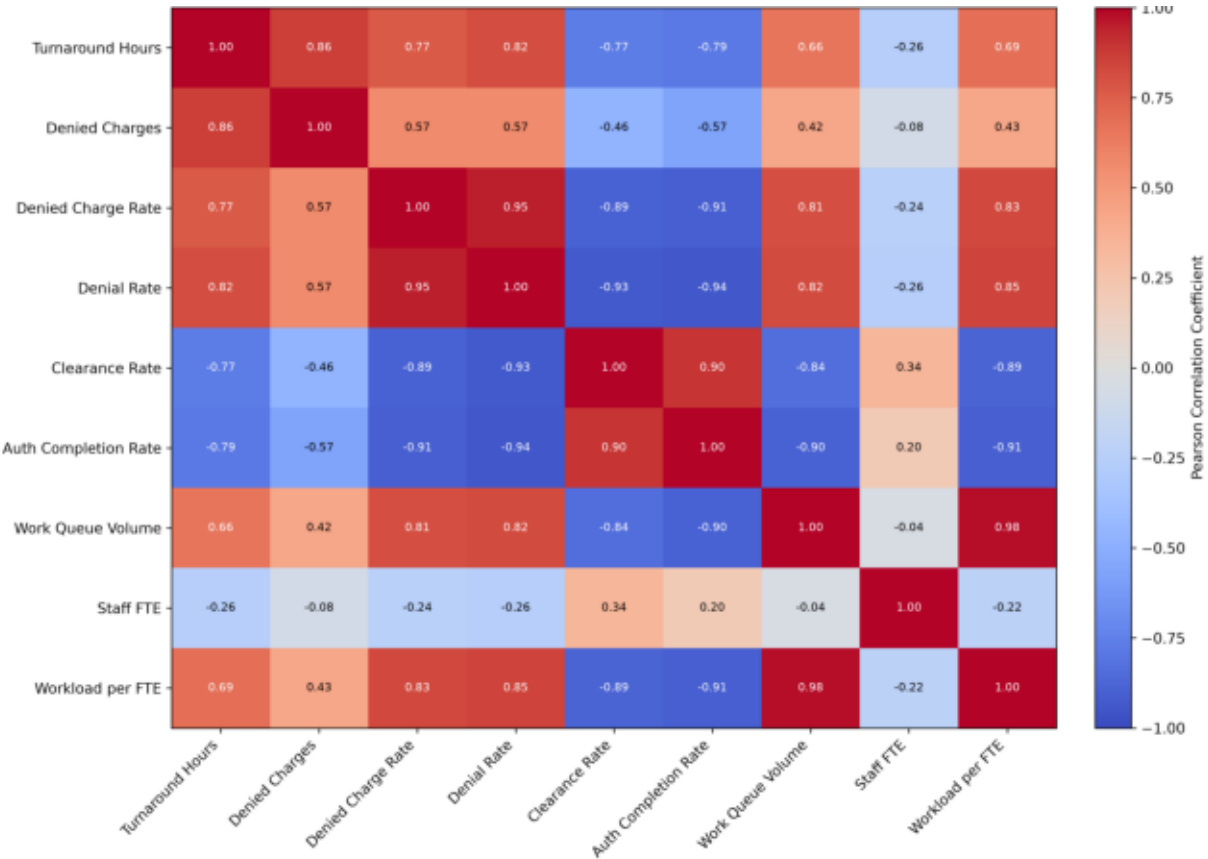
Reject H₀

Hypothesis decision

Primary Linear Regression Model



Pearson Correlation Matrix



Primary finding

- Strong positive relationship between turnaround time and denied charges.
- Pearson $r = 0.861$; $R^2 = 0.742$.
- Regression p-value < 0.001 .
- Within the synthetic dataset, each additional hour of average authorization turnaround time was associated with approximately \$12,495 in additional daily denied charges.

Recommended actions

- Prioritize requests nearing turnaround thresholds.
- Monitor payer and specialty bottlenecks.
- Align staffing with queue volume and workload.
- Use predictive alerts for earlier intervention.
- Validate the model using deidentified real-world operational data before production deployment.

Interpretation and limitations

- Synthetic dataset created for educational analysis.
- Results demonstrate association, not causation.
- Residual diagnostics indicate autocorrelation.
- Residuals depart from perfect normality.
- Future studies should use real-world multivariable data.

Business impact

- Supports proactive authorization management.
- Highlights denied-charge risk before service.
- Enables data-driven staffing decisions.
- Provides a reproducible analytics workflow.